

$$S(n, j) = G(n) - \sum_{i=1}^{j-1} f(p[i]) + \sum_{i=j}^{p[i]^2 \leq n} \sum_{e=1}^{p[i]^{e+1} \leq n} (f(p[i]^e) S(\lfloor \frac{n}{p[i]^e} \rfloor, i+1) + f(p[i]^{e+1}))$$